

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

Annexure A

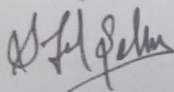
Scenario Based Assignment

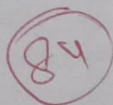
Code of Conduct:

I TamilSelvan Sasikumaran (192524124) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







Student 38: Tamil Selvan Sasikumar | Register Number: 192524134

1. Scenario

A firefighting robot is deployed inside a burning multi-storey building with an unknown internal layout. It must navigate safely to locate trapped occupants, avoid hazards (collapsed structures, fire zones), operate with limited visibility due to smoke, and make decisions without a complete floor plan. The robot updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search.
- ii. Analyze the challenges of partial observability and dynamic environments.
- iii. Describe how the robot builds and updates its internal model.
- iv. Compare online search with uninformed and informed search strategies.
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

2. Scenario

An airport is designing an AI system to automatically assign arriving flights to available gates. The system must satisfy multiple constraints: no two flights can occupy the same gate at overlapping times, a limited number of gates support wide-body aircraft, certain airlines have contractual gate preferences, ground crew availability constraints, and special accommodations for delayed connecting flights. The complexity increases as the number of flights and gates grows.

Tasks:

Formulate the problem as a Constraint Satisfaction Problem (CSP).

- i. Clearly define variables, domains, and constraints with explanation.
- ii. Explain how backtracking search works in this scenario.
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.
- iv. Analyze real-world challenges and justify the best strategy for scalability.

3. Scenario

A board-game company is developing an AI opponent for a turn-based strategy game similar to checkers, to be deployed on low-power tablets. The AI must compete against human players, make decisions within strict time limits, evaluate complex board states with multiple possible moves, and the decision tree is extremely large, making computation expensive on limited hardware.

Tasks:

- i. Explain how the Minimax algorithm models this problem.
- ii. Analyze the computational challenges of large game trees.
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.
- iv. Design an evaluation function and justify the factors considered.
- v. Recommend a strategy for balancing optimality and real-time performance.

Tasks:

Formulate this as an optimization problem and explain why traditional search is inefficient.

- Explain how Hill Climbing would work in this scenario and identify its limitations.
- Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
- Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
- Recommend the best approach and justify your choice based on scalability and adaptability.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 + 20 + 20 + 16 + 8$$

$$= 84$$

80

CO-2-AT-2

1) Fire fighting robot - online search

i) why online search?

The building layout is unknown and conditions change continuously. An online search agent can explore, learn, and make decisions in real time instead of requiring a complete map beforehand.

ii) Challenges of partial observability and dynamic environment

- Smoke reduces visibility.
- Sensors provide incomplete information.
- Fire and temperature can change.
- Structures may collapse or paths may become blocked.
- The robot must continuously adapt.

iii) How does the robot update its internal model?

The robot uses sensors to detect walls, obstacles, fire, and people. It creates a map and continuously updates it with newly discovered information and hazards.

iv) comparison of search strategies

Strategy - main feature

uninformed - No knowledge about the goal

informed - uses heuristic information

online - searches while exploring

online + informed - explores and uses useful information

v) Best approach

online informed search is most suitable because it supports real-time exploration, adapts to changing hazards and can select safer paths.

2) Airport Gate assignment - CSP

i) variables, domains and constraints

variables: Flights ($F_1, F_2, F_3, \dots$)

domains: Available gates ($G_1, G_2, G_3, \dots$)

constraints:

- No two flights use the same gate at the same time
- Aircraft must match gate requirement.
- Airline gate preferences should be considered.
- Ground crew must be available.

ii) Backtracking Search

Backtracking assigns a gate to each flight one by one. If a conflict occurs, it returns to the previous assignment and tries another gate.

iii) Constraint Propagation:

Constraint Propagation removes impossible gates from the domains. Forward checking detects conflicts early and reduces unnecessary searching.

iv) Best Strategy for Scalability:

Large airports have many flights, gates, delays, and constraints. Backtracking with constraint propagation, MRV, and forward checking provides better scalability.

3) Board - Game AI - minimax

i) How does minimax model the problem?

Minimax represents the game as a game tree. The AI tries to maximize its score, while the opponent tries to minimize it. The AI chooses the best move assuming the opponent plays optimally.

Computational Challenges

Game trees can become extremely large because every move creates several possible future moves. This increases time and memory requirements, especially on low-power devices.

iii) Alpha - Beta Pruning.

Alpha - Beta pruning removes branches that cannot affect the final decision. It gives the same result as minimax while examining fewer game states, making the search faster.

iv) evaluation function:

A simple evaluation function can consider:

- Number / value of pieces
- position of pieces
- available moves
- piece safety
- winning threats

v) Best strategy:

use depth-limited minimax with alpha - beta pruning, along with iterative deepening and a good evaluation function. This provides a balance between accuracy and real-time performance.